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Exploration on Experimental Economics and the Education Reform of "New Finance and Economics"

Yuanyuan Wang
Accounting School
Hebei University of Economics and Business
Shijiazhuang, China
e-mail: Wangyuanyuan201610@163.com

Shuyu Wang*
Business School
Hebei University of Economics and Business
Shijiazhuang, China
e-mail: 18601152336@163.com

Wenyue Li
Business School
Hebei University of Economics and Business
Shijiazhuang, China
e-mail: 15930796376@163.com

***Abstract*—The construction of new engineering, new medical science, new agricultural science and new liberal arts has provided a direction for the educational reform in universities and colleges in China, which are extended to the education of "new finance and economics" that will be a hot topic. Experimental economics, as a new discipline, has broken the previous inherent cognition of the theoretical research of finance and economics with the mode of experimental operation. In this paper, the relationship between experimental economics and education of "new finance and economics" will be explored from two dimensions: the relationship between experimental economics and the construction of New Finance and Economics courses and the relationship between experimental economics and students' innovation and entrepreneurship under the background of new finance and economics education. In the new era, the education of finance and economics needs innovative talents who combine theoretical knowledge with practical experience. It is found that the structure and specialty of the finance and economics major have been improved by integrating the experimental economics method into the courses, which has a far-reaching impact on the horizontal and vertical development of the financial and economic education reform.**

Keywords—"new finance and economics" education, experimental economics, creative talents

I. Background of "New Finance and Economics" Education Reform

The "Four News" construction strategic plan for the all-round development of new engineering, new medical science, new agricultural sciences and new liberal arts, which was written in the central document in 2008, will promote the level of discipline structure in colleges and universities in a point-to-area way, thus forming a first-class undergraduate professional cluster covering all disciplines with Chinese characteristics and world level, with an aim to keep students busy, keep education alive, and keep management tight.

New engineering focuses on training talents according to the "new requirements of engineering". New medical sciences are the reform of teaching system focusing on social reality. New agricultural sciences are the transformation and upgrade of disciplines based on the study of Xi Jinping's idea of ecological civilization and by means of science and technology. New liberal arts are the philosophical development that needs to integrate the new era ideas.

Correspondingly, at present, 612 new engineering research and practice projects have been identified for the first batch; 74 affiliated hospitals of colleges and universities have been established as the first batch of national demonstration centers for clinical teaching and training; the layout of new agricultural-related disciplines has been laid out; and the school of philosophical and social sciences with Chinese characteristics has been advocated.

According to the "Four News" reform plan of the Ministry of Education, the concept of "New Finance and Economics" education was put forward by Hebei University of Economics and Business and affirmed by the Ministry of Education. "New finance and economics" is the junction of education, economy and technology in the new era and is in a new historical position with unprecedented great changes in a century. However, the goal of the education reform of "new finance and economics" has not been finalized, and the ways and methods of the reform are also unclear. Therefore, it is worth considering and exploring how to grasp the opportunity of "four news" construction and carry out innovative reform of financial education in Colleges and universities according to the original characteristics of financial discipline.

II. Characteristics of Experimental Economics

What is studied in experimental economics is the real "actor", which provides data source for economic research, with the core content of quantitatively measuring the change trend of the variables studied by constructing a controllable micro-economic environment in the laboratory and controlling the necessary variables.

As a new discipline, experimental economics has unparalleled advantages over other disciplines. First of all, experimental economics brings the boring theory of economics to the laboratory for simulation, which enables students not only to conduct in-depth analysis of economic theory, but also to feel the development of economic laws, and thus better grasp the previous views. Secondly, through practical operation, experimental economics can well inspire students' innovative thinking, and innovation is the driving force of development and the key aspect of talent training.

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Finally, the teaching process of experimental economics involves the study and discussion of the experimental process, which enhances the interaction between teachers and students, and between students and students. Thus it is clear that experimental economics is undoubtedly an effective way to change the teaching mode of teachers (Fig. 1 and Fig. 2). All these characteristics make experimental economics a sustainable contribution in both teaching and scientific research [1].

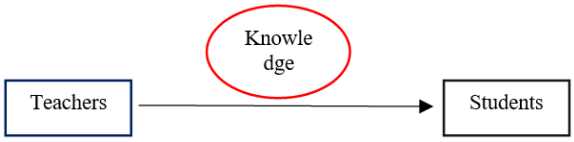

Fig.1. Traditional teaching mode

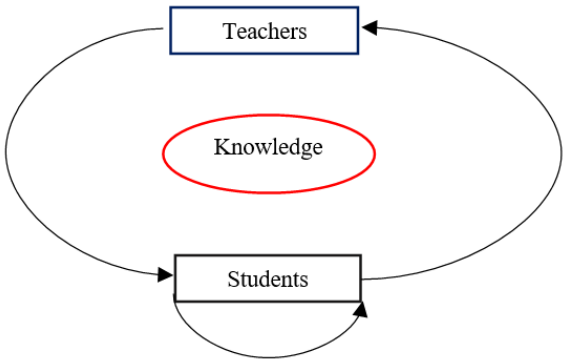

Fig. 2. The teaching mode based on the experimental economics

In the frontier of international academics, experimental economics has become increasingly recognized with the acceptance and high frequency publication of articles on experimental economics or research using experimental methods by a large number of leading economic journals. In domestic higher education, some first-class schools have built or are preparing laboratories of experimental economics, and the establishment of projects using experimental economics methods funded by national projects shows a rapid growth trend.

III. Relationship between Experimental Economics and New Finance and Economics Education

A. Relationship between Experimental Economics and the Construction of "New Finance and Economics" Course

1) Experimental economics helps to train all-round personnel with big data under the new finance and economics education system

Generally, Accounting, Finance and Economics are offered in financial and economic colleges and universities in China. Unlike traditional manufacturing and service industries, the financial sector places greater emphasis on the application of large data resources and the role of information flow. However, the courses of major financial colleges and universities in China are difficult to meet the social needs, due to the lack of scientific personnel training positioning on the one hand, and the lack of systematic knowledge of financial theory, professional project practice on the other [2].

If experimental economics is included in the pilot course of "new finance and economics" education reform taking experiments as the hitting point, then the positioning standards of financial and big data talents can be clearly defined, and the theoretical knowledge system of financial and economics courses can be more systematically summarized, while bringing professional project practice to students.

The integration of experimental economics can not only improve the systematic teaching, but also increase the teaching characteristics and teaching practice of finance and economics major in colleges and universities.

2) Experimental economics helps to train innovative talents under the new financial and economic education system

All along, innovative talents are one of the training objectives of domestic colleges and universities, and also one of the development needs of all walks of life [3]. Even under the changing political and economic environment in the world, the supply of innovative financial talents is still at a low level. However, colleges and universities have imperfect training modes for innovative talents in finance and economics, as well as teaching contents that are difficult to meet the current needs of social and economic development. Most schools only pay attention to the theoretical rules on textbooks, and do not put much effort into practical and hands-on links involving personal experience. As a result, students can not have the opportunity to experience actually, making it difficult for abstract book contents to be converted into ideological knowledge, and also killing students' enthusiasm for learning.

To a certain extent, practical activities are the soul of higher financial education. Therefore, the reform of new finance and economics education must focus on the practice and training of innovative talents. Experimental economics, exactly, provides students with good practice opportunities. In any case, experimental economics can enhance the students' awareness of innovation, activate their innovative thinking, and then cause them to think deeply about theoretical knowledge, or even resonate with each other, and finally carry out self-awareness and summary from a new perspective.

3) Experimental economics widens the contents of teaching reform under the new finance and economics education system

In the era of information technology, traditional teaching models can no longer meet the needs of the current industry. In the teaching reform under the educational system of "New Finance and Economics", the multimedia technology should be applied to the teaching content, and the resources on the Internet should be used reasonably to broaden the teaching content of students.

The experimental Economics Laboratory can be regarded as the practical training center of economics, and most of the courses involve computer operation, which can not only make reasonable use of multi-media technology, but also integrate Internet resources into teaching. Such a practical training center is more close to reality, and is conducive to the cultivation of vocational quality and professional skills of students.

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B. Relationship between Experimental Economics And Students' Innovation and Entrepreneurship under the Background of New Finance and Economics Education

1) Helpful to cultivate students' awareness of innovation and entrepreneurship

According to Opinions on Vigorously Promoting Innovation and Entrepreneurship Education in Institutions of Higher Education and Independent Entrepreneurship of Undergraduates issued by the Ministry of Education, vigorously promoting innovation and entrepreneurship education in Institutions of higher education is of great practical significance and long-term strategic significance for promoting scientific development of higher education, deepening education and teaching reform, and improving the quality of talent cultivation. However, finance and economics majors in colleges and universities are confronted with a series of problems in the practice of innovation and entrepreneurship education, such as weak awareness of innovation and entrepreneurship among college students, and unreasonable construction of innovation and entrepreneurship course system. Fortunately, the teaching mode of experimental economics provides students with a transaction platform which is suitable for reality and simulates reality, and enables them to truly experience the actual market conditions of innovation and entrepreneurship, which helps to enhance their interest in innovation and entrepreneurship.

2) Helpful to the innovation of "Internet +" teaching mode

With the rapid development of information technology in recent years, the teaching mode of "Internet +" has attracted more and more attention of teachers and students. In addition, in response to the development of the situation, the national and provincial education authorities have successively issued a series of documents, requiring the combination of scientific and technological revolution and higher education, to play the role of "Internet +" teaching mode, and to change the relatively weak teaching situation in general class [4]. Experimental economics, with its characteristics of computer operation and scientific experiments, has actively responded to the call of "Internet +" teaching. While using the network teaching mode, it has innovated the teaching class of Finance and economics [5].

TABLE I. INTRODUCTION OF MAJOR ECONOMICS LABS IN CHINA

Universities	Year	Introduction
Laboratory of Experimental Economics, Dongbei University of Finance and Economics	2009	With an area of 160 square meters, including lead block, experimentation area and control office area, and research direction of financial science, individual behavior and decision-making, policy wind tunnel, it is designed to promote experimental economics and its application in scientific research, provide experimental methods and technical support for other relevant scientific research institutions in the school, and carry out cooperation with major domestic economic management science laboratories.
Laboratory of Experimental Economics and Finance, Xiamen University	2009	The laboratory has excellent conditions, uses ORSEE to recruit experimental participants, and has a library of experimental participants composed of over 1,200 students from Xiamen University. As the first key laboratory of economic disciplines across the disciplines of Arts and Sciences in China, it is of great significance to establish the leading position of the economics disciplines of Xiamen University.
Selten Economics Lab, Nanjing Audit University	2018	With first-class economic experimental equipment and space, mainly researching the frontier issues related to social economy and politics in China, the laboratory is designed to build an exchange platform that can integrate related academic resources at home and abroad, in Jiangsu Province and within the university.
Laboratory of Economic Organization and Economic Behavior, Renmin University of China	2006	As the National Experimental Teaching Demonstration Center of Renmin University of China and one of the sub-laboratories of National Virtual Simulation Experimental Teaching Center of Renmin University of China, it provides the basic environment for economic experiments for teachers, students and scholars outside the university. By the end of 2013, 4 SCI/SSCI and more than 30 CSSCI papers have been published by researchers from this laboratory (up to 2011).
Laboratory of Experimental Economics, Shanghai University of Finance and Economics	2006	As a cooperation platform for economic, management science and other social science workers, the laboratory provides conditions for answering the current hot issues in China. A number of articles have been published in international top journals by researchers from the laboratory.
Key Laboratory of Economic Behavior Science, South China Normal University	2014	Including a behavioral laboratory and a neuro-decision laboratory, which are jointly sponsored by the Economics and Psychology disciplines of South China Normal University, as the Key Laboratory of Philosophy and Social Sciences in Guangdong Province, it mainly studies the economic decision-making behavior of micro-individuals, the macroeconomic and social effects of individual behaviors, and the evaluation of economic policies based on field experiments or quasi-natural experiments.
Experimental Center for Behavioral Science Research, Wuhan University	2016	With behavioral science and experimental methods as main research means and multi-platform integrated test management and data collection system as feature, an open experimental platform is constructed to promote transformation and innovation of research methods in related disciplines, enhance collaboration and cooperation with other domestic colleges and universities, and accelerate pace with international peers.

IV. CONSTRUCTION STATUS OF ECONOMICS LABORATORY

Experimental economics has become a cutting-edge discipline in the world, involving more research and covering a wide range of areas, and has been applied to some non-traditional economics fields. Moreover, the domestic relevant researches are also increasing rapidly. From the financing situation of national projects, there are more and more cross-studies involving experimental economic methods. According to the annual PhD thesis, more and more PhD graduates use experimental economic methods for project research. In addition, in recent years, economic laboratories have been built and put into use in many universities in China (see Table 1). Emerson and Taylor found through the experiments that incorporating experimental economics into classroom teaching yielded significant gains on the understanding examination of university economics [1].

V. CONCLUSIONS AND ENLIGHTENMENT

"New liberal arts" aims to develop the advanced culture of socialism with Chinese characteristics and the

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philosophical system of the new era, to promote the development of philosophical and Social Sciences in the new era, and to enhance the cultural soft power of our country. "New Finance and Economics" education has emerged as the times require, providing hope for the reform of Finance Specialty in Chinese colleges and universities. Because of its frontier discipline and innovative practicality, experimental economics has become the most important part of the new finance and economics education and reform. In recent years, the method of experimental economics has been widely used in many fields, such as microeconomics, finance, financial management, etc., which has been used together with the method of big data acquisition and analysis to test and analyze simultaneously in many disciplines. Practical and integrated experimental economics can not only improve students' own knowledge structure level, but also stimulate their subjective initiative, thus cultivating innovative financial talents, providing new ideas and methods for "new finance and economics" education, which is an effective measure to improve the quality of teaching and ensure the positive interaction between teaching and research.

In order to meet the needs of current national construction, each major of each university should establish a sound system, carry out overall planning, actively seek development and transformation, pay attention to the construction of discipline connotation and discipline quality, and promote the integration of disciplines and specialties as a whole. The financial education in the new era needs innovative talents with the combination of theoretical knowledge and practical experience. The integration of experimental economics into the courses of Finance and economics has optimized the structure and professionalism of the disciplines, and has a far-reaching impact on the horizontal and vertical development of the reform of Finance and economics education.

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